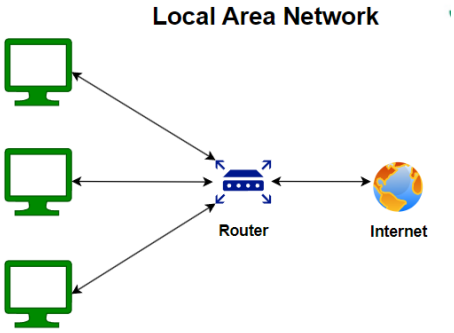
UNIT 2

Network Infrastructure

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1. LAN (Local Area Network)

A Local Area Network (LAN) connects devices within a limited area, like an office, building, or even a home. It's designed to cover short distances, allowing all connected devices to share resources such as files, printers, and internet access. In most LANs, speeds range from 10 Mbps to 100 Mbps, though faster speeds are now common. The typical setups, or topologies, for LANs include bus, ring, and star formations.



Key Features of a LAN Network

A Local Area Network (LAN) has several key features that make it an essential part of homes, offices, and schools. LANs connect devices like computers, printers, and smartphones within a limited area, such as a building or campus. Here are the main features of a LAN:

- Limited Coverage: A LAN only covers a small geographic area, typically within a single building or location.
- High Data Transfer Speeds: LANs offer fast data transfer rates, ranging from 10 Mbps to 100 Mbps, and even higher speeds in modern networks.
- Resource Sharing: Devices connected to a LAN can easily share resources like files, printers, and internet connections.
- Secure and Controlled Access: LANs are more secure as they are private networks, allowing for better control over who can access the connected devices.

How LAN Works

A **Local Area Network (LAN)** works by connecting multiple devices within a small area, like an office or home, allowing them to communicate and share resources. Here is how a LAN operates:

1. **Connection of Devices**: Devices such as computers, printers, and smartphones are connected to the network through cables (Ethernet) or wirelessly (Wi-Fi). Each device is equipped with a Network Interface Card (NIC) that allows it to connect to the LAN.
2. **Data Transmission**: When one device sends data, like a file, it travels through the network. If it's a wired LAN, the data moves through Ethernet cables, while in a wireless LAN, it's transmitted via radio waves.
3. **Switches and Routers**: A switch helps direct the data to the right device within the LAN, ensuring efficient communication. If internet access is needed, a router connects the LAN to the internet, allowing devices to go online.
4. **Resource Sharing**: Once connected, devices can easily share resources, like files, printers, or an internet connection, making it simple and fast for users to work together.

Benefits of a LAN

A Local Area Network (LAN) offers several advantages, making it an essential part of homes, offices, and schools. Here are the key benefits of using a LAN:

Fast Data Sharing: LANs allow connected devices, like computers and printers, to share files and resources quickly. Whether you're sending documents, streaming videos, or accessing shared drives, LAN speeds make everything faster and more efficient.
Cost-Effective: Once set up, a LAN reduces the need for extra devices or duplicate resources. For example, multiple computers can share a single printer or internet connection, saving both money and space.
Improved Communication: With LANs, devices can communicate directly with each other. This is useful for tasks like file transfers, video calls, or collaborative work, making teamwork easier and more productive.
Centralized Data Management: LANs enable centralized storage, where files can be saved in one location and accessed by everyone in the network. This helps with data organization and ensures easier backups.
Enhanced Security: Since a LAN is a private network, it's easier to control access and keep unauthorized users out. You can implement security measures like firewalls and passwords to protect sensitive data.
Reliable and Stable: Wired LANs, in particular, are known for their stability and reliability. They are less likely to experience interference, making them ideal for environments where uninterrupted connectivity is crucial.

History of LAN

LAN (Local Area Network) has come a long way since its early days, and its journey is fascinating.

1973: The Birth of Ethernet

Dr. Robert M. Metcalfe invented Ethernet to connect computers within a building and link them to the first Xerox laser printer. He named it "Ethernet" after the concept of Aether, the invisible medium once thought to conduct light. The Ethernet worked by sending data through a coaxial cable, allowing

computers to communicate. It used a collision-detection method—if two devices sent data at the same time, they would pause and try again after a short random wait.

1977: First Commercial LAN

The first commercial LAN was set up by Datapoint Corp. at Chase Manhattan Bank. Unlike Ethernet, this LAN used a "token-passing" system, where devices took turns transmitting data, ensuring no collisions.

1979: Ethernet Goes Public

Metcalfe launched 3Com to make Ethernet products available to the public, marking the start of widespread LAN use.

1985: Ethernet Becomes the LAN Standard

The IEEE (Institute of Electrical and Electronics Engineers) made Ethernet the official standard for LAN, giving it a significant boost in popularity.

1990: Ethernet Wins the LAN Battle

Ethernet became the dominant LAN technology by switching to a twisted-pair cable, which made it faster and reduced interference. It outperformed other LAN systems, making it the preferred choice for networks.

1991: Work Begins on Wireless LAN (WLAN)

The IEEE began working on wireless LANs, inspired by an earlier wireless prototype called ALOHA. This set the stage for future wireless technologies.

1997: The Birth of Wi-Fi

In 1997, the IEEE introduced the 802.11 standard, better known as Wi-Fi. Unlike Ethernet's collision-detection, Wi-Fi uses a method called CSMA/CA (Carrier Sense Multiple Access/Collision Avoidance), which ensures that devices wait for the network to be idle before transmitting data.

2. Ethernet (IEEE standard 802.3) LAN

Ethernet is a set of technologies and protocols that are used primarily in LANs. It was first standardized in the 1980s by IEEE 802.3 standard. IEEE 802.3 defines the physical layer and the medium access control (MAC) sub-layer of the data link layer for wired Ethernet networks.

Ethernet is classified into two main categories: classic Ethernet and switched Ethernet. Classic Ethernet is the original form providing data rates between 3 to 10 Mbps, commonly referred to as 10BASE-X variants. Switched Ethernet uses switches to connect stations in the LAN, replacing repeaters and allowing full bandwidth utilization.

IEEE 802.3 Popular Versions

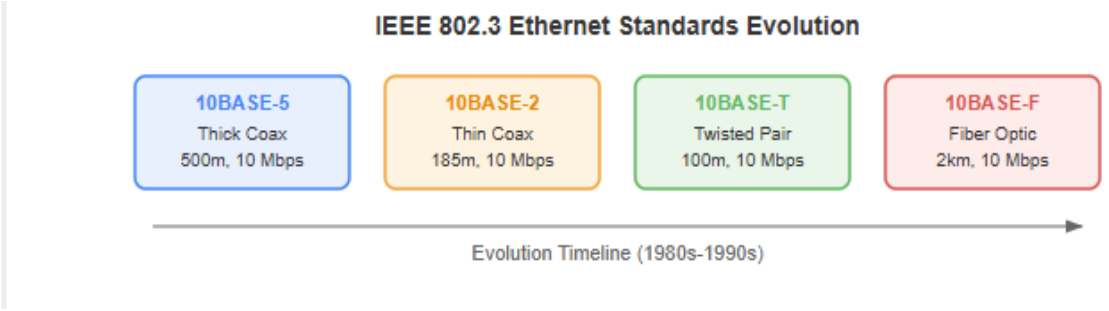
The most significant versions of IEEE 802.3 protocol include:

IEEE 802.3 (10BASE-5) − The original standard using thick coaxial cable. Connections are made by drilling into the cable to reach the core. Maximum throughput is 10 Mbps with baseband transmission and maximum segment length of 500m.

IEEE 802.3a (10BASE-2) − Standard for thin coax where cable segments are connected by BNC connectors. The "2" refers to the maximum segment length of approximately 200m (185m precisely).

IEEE 802.3i (10BASE-T) − Standard for twisted pair using unshielded twisted pair (UTP) copper wires. Further variations include IEEE 802.3u for 100BASE-TX, 100BASE-T4, and 100BASE-FX.

IEEE 802.3j (10BASE-F) − Standard for Ethernet over fiber optic cables as the transmission medium.



Frame Format Comparison

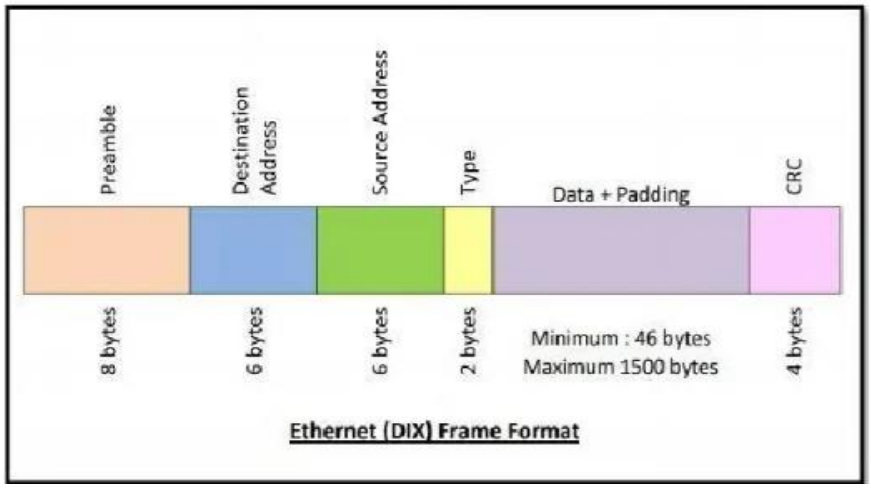
The frame formats differ slightly between Classic Ethernet and IEEE 802.3:

Field	Classic Ethernet	IEEE 802.3	Purpose
Preamble	8 bytes	7 bytes	Synchronization and timing
Start Frame Delimiter	Part of preamble	1 byte	Frame start indication
Destination Address	6 bytes	6 bytes	Target station MAC address
Source Address	6 bytes	6 bytes	Sender station MAC address
Length/Type	Type (2 bytes)	Length (2 bytes)	Protocol type or data length
Data + Padding	46-1500 bytes	46-1500 bytes	Payload with minimum padding
CRC	4 bytes	4 bytes	Error detection

Key Frame Fields

- **Preamble** − Provides synchronization pulses for transmission timing
- **Start of Frame Delimiter** − Contains alternating pattern ending with two consecutive ones
- **Address Fields** − 6-byte MAC addresses for source and destination stations
- **Data Field** − Variable length (46-1500 bytes) carrying upper layer data
- **CRC** − 4-byte cyclic redundancy check for error detection

Frame Format of Classic Ethernet IEEE 802.3



Frame Format of Classic Ethernet 802.3

Classic Ethernet frames can be either of Ethernet (DIX) or of IEEE 802.3 standard. The frames of the two standards are very similar except for one field. The main fields of a frame of classic Ethernet are:

- **Preamble** – It is the starting field that provides alert and timing pulse for transmission. In case of Ethernet (DIX) it is an 8 byte field and in case of IEEE 802.3 it is of 7 bytes.
- **Start of Frame Delimiter (SOF)** – It is a 1 byte field in an IEEE 802.3 frame that contains an alternating pattern of ones and zeros ending with two ones.
- 11000011 DATA11000011
- **Destination Address** – It is a 6 byte field containing physical address of destination stations.
    - **Source Address** – It is a 6 byte field containing the physical address of the sending station.

- **Type/Length** – This is a 2 byte field. In case of Ethernet (DIX), the field is type that instructs the receiver which process to give the frame to. In case of IEEE 802.3, the field is length that stores the number of bytes in the data field.
- **Data** – This is a variable sized field carries the data from the upper layers. The maximum size of data field is 1500 bytes.
- **Padding** – This is added to the data to bring its length to the minimum requirement of 46 bytes.
- **CRC** – CRC stands for cyclic redundancy check. It contains the error detection information.

The generally uses specifications of IEEE 802 are:

- **IEEE 802.3 (Ethernet)**

  The IEEE 802.3 standard determines the CSMA/CD access control protocol. The best known scheme for controlling a local area network on a bus structure is carrier sense multiple action with collision detection (CSMA/CD).

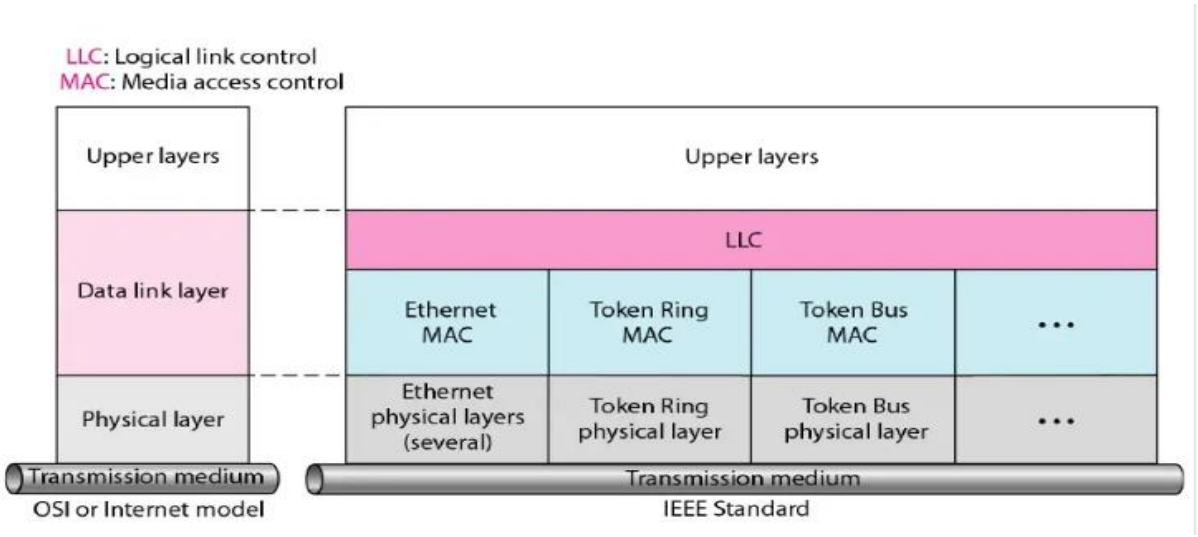
- **IEEE 802.4 (Token Bus)**

  IEEE 802.4 describes a token bus LAN standards. In token passing methods, stations connected on a bus are arranged in a logical ring. In this method only the station having token (token holder) is being permitted to transmit frames.

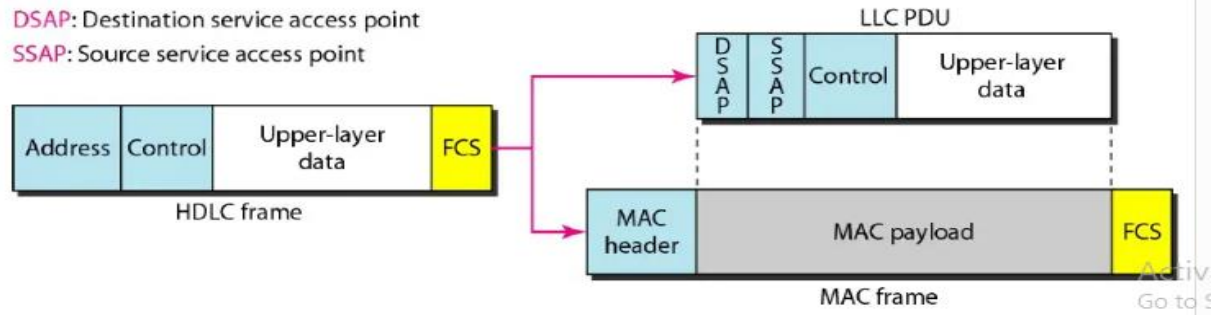
- **IEEE 802.5 (Token Ring)**

  IEEE 802.5 describes the token ring standards. In a token ring a special bit pattern, called the token, circulates around the ring whenever all stations are idle. The sequence of token is determined by the physical locations of the stations on the ring.

IEEE Standards for LAN



HDLC frame compared with LLC and MAC frames



Ethernet

Thick Ethernet

- Thick Ethernet was the first commercially available form of cabling supported by Ethernet.
- It is technically known as 10-BASE-5.
- Here, 10 is the maximum throughput, i.e. 10 Mbps, BASE denoted use of baseband transmission, and 5 refers to the maximum segment length of 500 metres (1,600 ft).
- This type of cabling allows 100 stations to be connected to it by vampire taps.
- The stations share a single collision domain.

Thin Ethernet

- Thin Ethernet, popularly known as thinnet, is among the family of Ethernet standards that uses thinner coaxial cable as a transmission media.
- It is technically known as 10-BASE-2.
- Here, 10 is the maximum throughput, i.e. 10 Mbps, BASE denoted use of baseband transmission, and 2 refers to the maximum segment length of about 200 metres (precisely 185 metres).
- This type of cabling allows a maximum of 30 stations to be connected to it by BNC connectors with 50 centimetres minimum gap between subsequent stations.

Features of Cable and Network

The salient features of 10-BASE-2 Ethernet cabling are -

- 10-BASE-2 use RG-58 A/U coaxial cable.
- It is thinner, more flexible, more economic and easier to install than the coaxial cable used in thick Ethernet.
- The cable has 10 Mbps transmission speed.
- The maximum segment length is 185 m and the minimum gap between stations is 50 cm.
- The maximum number of stations that can be connected is restricted to 30.

- Thinnet uses Manchester coding.
- A low-to-high transition in the middle of the bit period is encoded as binary 0 while a high-to-low transition in the middle of the bit period is encoded as binary 1.
- It uses BNC T-connector for connecting with the stations network interface card (NIC) and also for joining cables.

Ethernet Throughput

- Throughput of a system refers to the rate of processing of a task there by generating results.
- Ethernet is a set of technologies primarily used in LANs, whose primary data units are frames.
- The throughput of Ethernet is measured by the rate of successful delivery of frames over a communication channel.

Ethernet IEEE 802.3

- Ethernet is a set of technologies and protocols that are used primarily in LANs.
- It was first standardized in 1980s by IEEE 802.3 standard.
- IEEE 802.3 defines the physical layer and the medium access control (MAC) sub-layer of the data link layer for wired Ethernet networks.

3. WAN

A WAN (Wide Area Network) is to connect multiple smaller Local Area Networks (LANs). It is a computer network designed. WANs can help in communication, the sharing of information, and much more between systems or devices from around the world through a WAN provider.

WAN stands for Wide Area Network. It is a computer network that covers a large geographical area consisting of two or more LANs or MANs. These networks are established with leased telecommunication circuits, in which two sides which are connected have routers that connect the LAN of both sides together in a network to facilitate communication.



History of WAN

The roots of WAN are connected to the U.S Department of defense which developed ARPANET to let researchers communicate and share computer resources remotely. The connection can be circuit-switched telephone lines, radio wave transmission or optical fiber transmission.
It is used to exchange data with users all over the world, they can be client, employee, buyer, seller, student, etc. WAN has the ability to transmit data, image, audio data, and video data over large distances.

Software-Defined WAN (SD-WAN)

- It is a technique for making WAN architectures easier to construct, run, and administer is software-defined WAN (SD-WAN). It relies on virtualization, overlay networks, application-level policies and onsite SD-WAN devices and software platforms.
- SD-WAN improves the efficiency of data transfer across a WAN by shifting traffic to less expensive network links to replace more expensive leased or MPLS lines.

Types of WAN Technologies

There are mainly two technologies that are used in the WAN network design.

- **Circuit switching:** Circuit switched networks operate on the virtual connection principle, which dictates that all messages will take the same way and that resources along this path are set aside for this connection.

- **Packet Switching:** The size of a packet in a packet switched network is dictated by the outgoing link, and these packets may follow different route. These packets are ready to collected and reassembled at the destination.

- **TCP/IP protocol suite:** TCP/IP is a protocol suite of foundational of the internet protocols used to interconnect devices on Internet and other computers networks or device network. Full form of TCP/IP is Transmission Control Protocol/Internet Protocol.

- **Router:** A router is a networking device which transfers data packets between device networks and also we can say it is used to interconnect LANs to form a wide area network (WAN).

- **Packet over SONET/SDH (PoS**): Packet over SONET and SDH is a communication protocol used for WAN transport. When using optical fiber and SONET or SDH communication protocol used to defines how point-to-point links communicate.

- **Multiprotocol Label Switching (MPLS):** Multi Protocol Label Switching (MPLS) is an IP packet routing technique and also a network routing optimization technique that routes IP packet through paths via labels instead of looking at complex routing tables of routers.

Characteristics of WAN

•**Broader Reach:** The reach of WAN in terms coverage of geographical area is very high which can be a region, country or the world itself.

•**Higher Capacity:** The capacity of WAN in terms of number of LANs or WANs connected in a network is very high, which results in connection of large number of user over different location all around the globe.

•**Use of Public Carrier:** WAN uses telephone network, cabled system, satellites etc for connection and transmission purpose which are easily available.

•**Resource Sharing:** WAN enables its users to share data and information over large area. Computer resources can be accessed remotely which makes transmission and exchange of data very easy.

Advantages of WAN

- It covers large geographical area which enhances the reach of organisation to transmit data quickly and cheaply.
- The data can be stored in centralised manner because of remote access to data provided by WAN.
- The travel charges that are needed to cover the geographical area of work can be minimised.
- WAN enables a user or organisation to connect with the world very easily and allows to exchange data and do business at global level.

Disadvantages of WAN

- Traffic congestion in Wide Area Network is very high.
- The fault tolerance ability of WAN is very less.
- Noise and error are present in large amount due to multiple connection point.
- The data transfer rate is slow in comparison to LAN because of large distances and high number of connected system within the network.

4.Internet

Internet and networking are important parts of the modern age, providing various conveniences in personal and commercial life. The Internet is a vast network connecting computers, servers, and mobile devices worldwide.

Through it, users can share information, communicate, and access various services. Networking, which includes technologies like LAN, WAN, and WLAN, helps organizations and individuals share their resources and collaborate.

With the use of the Internet, ideas, information, and tools can be easily exchanged, creating new opportunities in education, business, and entertainment. Thus, the Internet and networking play a crucial role in our social and economic development.

Definition and Uses of the Internet

The Internet is a large global network of interconnected computer networks that uses the Internet Protocol Suite (TCP/IP) to facilitate communication and data exchange between

devices. It serves as a platform for various forms of communication, including email, social media, file sharing, and web browsing.

Definition:

- **Global System:** The Internet is a collection of millions of private, public, academic, business, and government networks connected together.
- **Protocol Suite**: It mainly relies on Transmission Control Protocol (TCP) and Internet Protocol (IP) to ensure reliable communication and data transfer.
- **Interconnectivity:** The Internet allows different types of devices and networks to connect and communicate regardless of their internal technologies.

Uses:

- **Communication:** Provides instant messaging, video calls, and emails, enabling real-time communication worldwide.
- **Information Access**: Provides access to vast amounts of information via websites, databases, and online resources.
- **Social Networking:** Platforms like Facebook, Twitter, and Instagram allow users to connect with others and share content.
- **E-commerce:** Enables online shopping, banking, and financial transactions, transforming how businesses operate and consumers shop.
- **Education:** Offers online courses, webinars, and educational resources, making learning accessible to wider audiences.
- **Entertainment:** Streaming services, online gaming, and social media provide various forms of entertainment.

The Internet has become an inseparable part of modern life, influencing how we communicate, work, learn, and entertain ourselves.

Types of Networks (LAN, MAN, WAN)
Computer networks are classified based on their size, scope, and technology used. The three main types are Local Area Network (LAN), Metropolitan Area Network (MAN), and Wide Area Network (WAN). Here is a description of each type:

Local Area Network (LAN)
Definition: A LAN connects computers and devices within a limited geographic area, such as a home, school, or office building.
Features:
High data transfer rates.
Low latency.
Usually owned and managed by a single organization or individual.
Examples: Home Wi-Fi networks, office intranets.

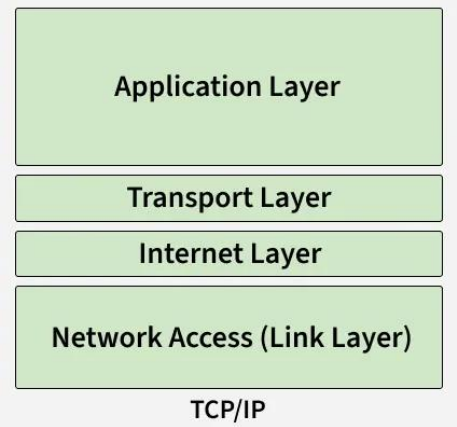
Metropolitan Area Network (MAN)
Definition: A MAN covers a larger geographic area than a LAN but is smaller than a WAN, usually spanning a city or a large campus.
Features:
Can connect several LANs within a specific area.
Often used to connect government buildings, universities, or large businesses.
Provides high-speed connections over medium distances.
Examples: City-wide Wi-Fi networks, cable television networks.

Wide Area Network (WAN)
Definition: A WAN covers a vast geographic area, often connecting multiple LANs and MANs across cities, countries, or continents.
Features:
Data transfer speeds are lower compared to LAN and MAN due to longer distances.
Usually owned by multiple organizations or service providers.
Uses different transmission media such as satellite links and leased telecommunication lines.
Examples: The Internet, corporate networks spread over multiple locations.

5. TCP/IP Reference Model

The TCP/IP model is a layered networking framework that explains how data is communicated between devices over a network using standardized protocols to ensure reliable and efficient transmission.

- Defined as a four-layer architecture consisting of Application, Transport, Internet, and Network Access layers.
- Standardized by RFC 1122, which specifies its structure and behavior.
- Simpler and more practical than the seven-layer OSI model.
- Serves as the core framework of the modern Internet and networking systems.

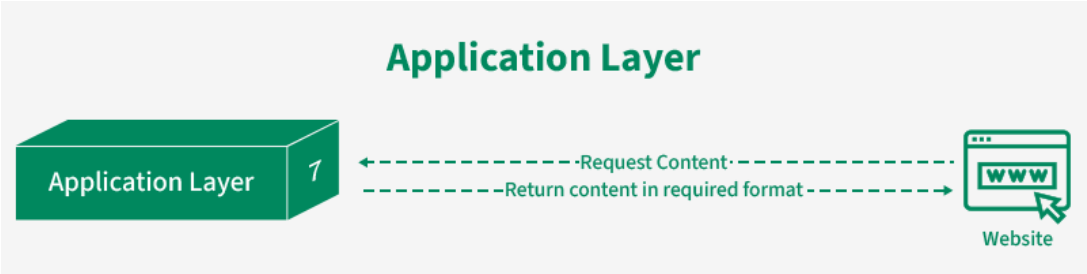


Layers of TCP/IP Model

1. Application Layer

This is the top layer of the TCP/IP model, where applications like web browsers, email clients, and file-sharing tools interact with the network.

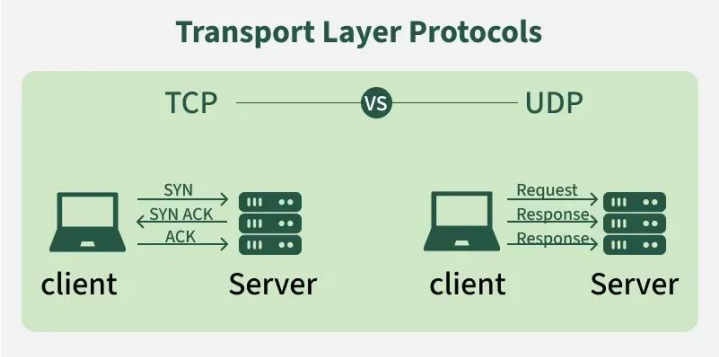
- Acts as a bridge between user applications and lower network layers.
- Supports protocols such as HTTP, FTP, SMTP, and DNS.
- Handles data formatting so information is correctly understood by both sender and receiver.
- Provides encryption for secure communication.
- Manages sessions to track ongoing connections.



2. Transport Layer

Ensures reliable and efficient delivery of data between devices, managing segmentation, ordering, and retransmission as needed.

- Breaks large messages into packets and reassembles them at the destination.
- TCP checks for errors, resends lost data, and ensures correct order.
- UDP provides low-latency, connectionless delivery without error checking.
- Prevents the receiver from being overwhelmed by regulating data flow.
- Uses port numbers to allow multiple applications to share the network simultaneously.



TCP (Transmission Control Protocol): TCP is used when reliability and accuracy are important. It ensures that data is delivered exactly as sent.

- TCP detects errors in the data using checksums to ensure integrity.
- If any data is lost or corrupted during transmission, TCP automatically resends it.
- Data is broken into packets, and TCP ensures these packets arrive in the correct sequence.
- TCP establishes a connection between sender and receiver before sending data, maintaining a stable session throughout the communication.
- Loading web pages, downloading files, sending emails, or any application where data must be complete and accurate.

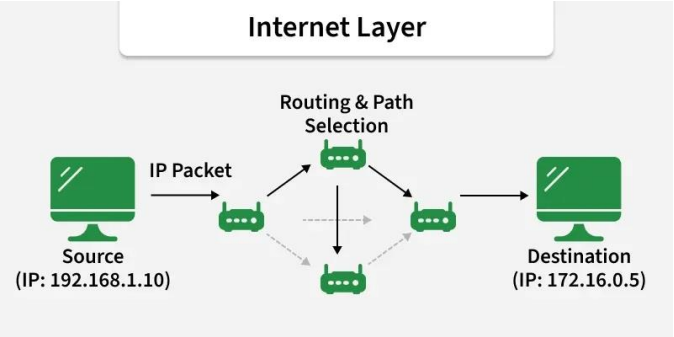
UDP (User Datagram Protocol): UDP is used when speed is more important than perfect accuracy. It is faster but does not guarantee reliable delivery.

- UDP detects errors via checksums but does not verify or recover from them.

- Lost or damaged data is not resent.
- Packets may arrive out of order, and the protocol does not fix it.
- Because it avoids extra checks and connections, UDP is faster and uses fewer resources.
- Live video streaming, online gaming, VoIP calls, or real-time applications where speed matters more than reliability.

3. Internet Layer

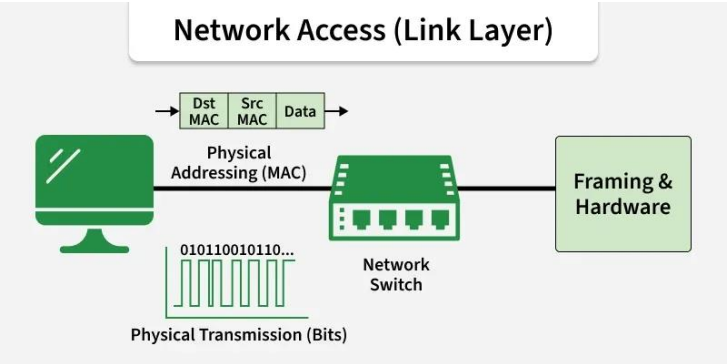
Responsible for addressing, packaging, and routing data packets so they can travel across networks and reach the correct destination device. It ensures that data can move between different networks efficiently.



- Assigns IP addresses to identify source and destination devices.
- Determines the best path for data to travel across networks.
- Breaks large packets into smaller ones for transmission and reassembles them at the destination.
- Primarily uses IP (Internet Protocol), along with supporting protocols like ICMP for error reporting and ARP for address resolution.

4. Network Access (Link Layer)

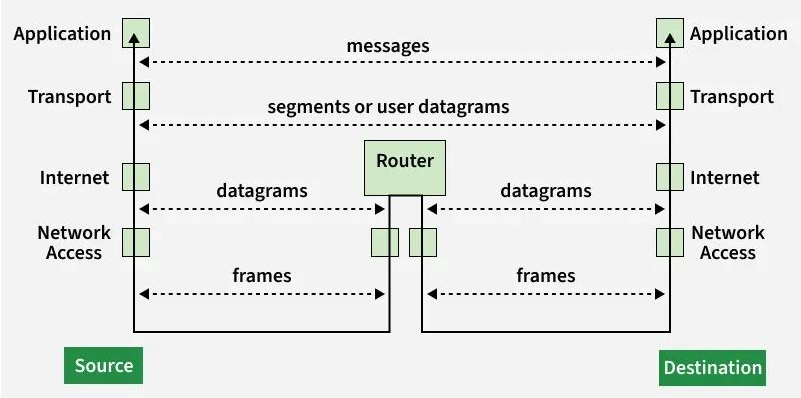
Responsible for physically transmitting data over network hardware, including cables, switches, and wireless connections. It handles how data is formatted for the network medium and ensures it reaches the next device on the path.



- Sends and receives raw bits over physical media like Ethernet cables, fiber optics, or Wi-Fi.
- Organizes data into frames for proper transmission and recognition by devices.
- Detects transmission errors using checksums or CRC.
- Uses hardware addresses to identify devices within the same network segment.
- Determines how multiple devices share the same physical medium, avoiding collisions.

Working

The working of TCP/IP can be explained with the help of the diagram given below and explained:



When Sending Data (From Sender to Receiver)

- **Application Layer:** The user's software (like a web browser or email client) creates the data and passes it to the next layer.
- **Transport Layer:** The data is broken into segments, and TCP or UDP adds control information to ensure reliable delivery or fast transmission.
- **Internet Layer:** Each segment is encapsulated into packets with IP addresses so it can be routed across networks to the destination device.
- **Network Access (Link) Layer:** The packets are converted into frames suitable for the physical network (Ethernet, Wi-Fi) and transmitted over cables or wireless signals.

When Receiving Data (At the Destination)

- **Network Access Layer:** The frames are received from the physical medium and checked for errors.
- **Internet Layer:** Frames are unpacked to extract packets and use the IP address to ensure it reaches the correct device.
- **Transport Layer:** Segments are reassembled into the original message, and any missing or corrupted data is corrected (if TCP is used).
- **Application Layer:** The complete data is delivered to the user application (like the browser displaying a webpage or the email client showing a message).

Advantages and Disadvantages of TCP/IP

Advantages

- **Widely Used:** Forms the foundation of the Internet and most modern networks.
- **Platform Independent:** Works on different hardware and operating systems.
- **Reliable Communication:** TCP ensures error checking, delivery confirmation, and data integrity.
- **Scalable:** Can support small networks to global Internet-scale networks.
- **Open Standard:** Free to use and not controlled by a single organization.

Disadvantages

- **Complexity for Beginners:** Can be difficult to fully understand all protocols.
- **No Strict Layer Enforcement:** Unlike OSI, layer boundaries are not rigid, which may cause implementation variations.
- **Overhead:** TCP's error checking and reliability features can add extra data overhead.
- **Security Limitations:** Basic TCP/IP was not designed with strong built-in security; additional protocols like TLS/SSL are needed.
- **Limited Multimedia Support:** Original design focused on data, not optimized for real-time audio/video (needs extra protocols).

6. Domain Name Server

Translation of human-readable domain names into IP addresses enables computers to communicate over the internet, a function performed by the Domain Name System (DNS), which operates in a hierarchical and distributed manner while improving performance through caching.

- Provides name-to-address resolution using a hierarchical system (Root, TLD, Authoritative servers) to locate domain information.
- Improves performance by using caching mechanisms, reducing lookup time for repeated requests.

Example: When you enter a website name like google.com in your browser, the system does not understand the name directly. DNS translates it into an IP address of the server hosting the website. This allows your browser to connect to the correct server and load the webpage.

Working
DNS process can be broken down into several steps, ensuring that users can access websites by simply typing a domain name into their browser.

- **User Input:** The user enters a domain name
- **Local Cache Check:** The browser or OS checks its cache for a stored IP address.
- **DNS Resolver Query:** If not found, the request is sent to a DNS resolver (usually by ISP).
- **Root Server Query:** The resolver queries a root server, which points to the correct TLD server.

- **TLD Server Response:** The TLD server directs the resolver to the domain's authoritative server.
- **Authoritative Server Response:** The authoritative server returns the actual IP address.
- **Final Response:** The resolver sends the IP back to the user, and the browser connects to the server.

Structure of DNS

The structure of DNS is hierarchical in nature, enabling scalable and organized domain name resolution across the global Internet.

1. Root

The topmost level of the DNS hierarchy.

- Represented by a dot (.) at the end of a domain name
- Acts as the starting point of domain resolution

2. Top-Level Domains (TLDs)

The level directly below the root that defines domain extensions.

- Includes extensions like .com, .org, .net, .edu
- Helps categorize domains by purpose or region

3. Second-Level Domains

The main domain name registered by an organization.

- Appears before the TLD (e.g., "example" in example.com)
- Uniquely identifies a domain under a TLD

4. Subdomains

Extensions of the main domain used for organization.

- Examples: www, mail, blog
- Helps structure different parts of a website

5. Hostnames

Identifies specific servers or devices within a domain.

- Examples: web1, mailserver, ftp
- Maps to actual IP addresses using DNS records

Responsible for storing DNS records and answering queries to resolve domain names into IP addresses, this function is performed by a Domain Name Server.

- Stores DNS records such as A, AAAA, MX, CNAME, NS, and PTR.
- Responds to queries from clients or other DNS servers.
- Operates within the hierarchical DNS structure.
- Can perform either recursive or authoritative functions.
- Enables communication between users and web servers by supplying necessary DNS data.

DNS Lookup

Also known as DNS Resolution, is the process of converting a domain name into its corresponding IP address so that a device can establish communication with the target server.

Key Components

The key components of DNS work together to resolve domain names into IP addresses efficiently.

1. DNS Resolver: Starts the DNS lookup process and acts as an intermediary between client and DNS servers.
- Initiates the DNS query from the client side.
- Forwards requests to appropriate DNS servers to get the IP address.

2. Recursive Query: A query where the resolver fetches the complete answer on behalf of the client.
- Resolver performs full lookup across multiple DNS servers if needed.
- Returns the final IP address or an error to the client.

3. Iterative Query: A query where the server provides the best information it has or a referral.
- Returns partial response or points to another DNS server.
- Helps the resolver move step-by-step toward the final answer.

4. Non-Recursive Query: A query where the answer is already available in cache or authoritative server.
- Response is returned immediately without further lookup.
- No additional DNS server communication is required.

DNS Caching
Technique that temporarily stores DNS records to reduce repeated queries and improve resolution efficiency.

- Stores previously resolved domain records locally.
- Reduces response time for future requests.
- Minimizes load on external DNS servers.
- Improves overall network performance.

TTL (Time-to-Live)
TTL defines the duration for which a DNS record is considered valid in cache.

- Specifies how long a DNS record remains stored in cache.
- Defined by the authoritative DNS server.
- Measured in seconds.
- After expiration, a new DNS lookup is required.

Example: If the TTL value is 3600 seconds, the record remains cached for one hour before it must be refreshed.

Reverse DNS Lookup
Reverse DNS Lookup is the process of resolving an IP address to its associated domain name, performing the opposite function of standard DNS resolution.

- Converts an IP address into a domain name.
- Uses special domains such as in-addr.arpa (IPv4) and ip6.arpa (IPv6).
- Relies on PTR (Pointer) records for mapping.

Reverse DNS is commonly used for:

1. Network Diagnostics

Helps in analyzing and troubleshooting network issues.

- Helps administrators identify the domain linked to a specific IP address.
- Assists in traffic analysis and troubleshooting.

2. Email Security

Ensures emails are sent from legitimate sources.

- Used by mail servers to verify sender authenticity.
- Helps reduce spam and fraudulent email activity.

DNS Record Types

Different DNS record types are used to store specific information about a domain.

- **A Record:** Maps a domain name to its corresponding IPv4 address.
- **CNAME Record:** Creates an alias that points one domain name to another domain name.
- **MX Record:** Specifies the mail server responsible for receiving emails for a domain.
- **TXT Record:** Stores text information used for verification and email authentication purposes

Internet Industry Structure

Internet infrastructure is a key enabler of e-business. The infrastructure consists of the following players intermediating between end users and content publishers:

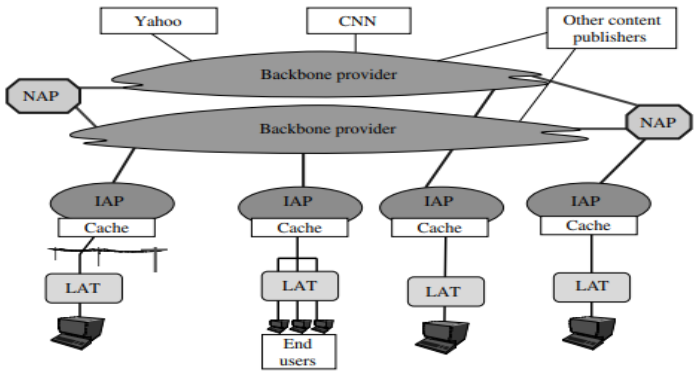
1. Internet access providers (IAPs) such as AOL and Earthlink that provide retail-level Internet access to the end users.
2. Local area transport (LAT) service providers such as local phone companies and cable franchises that connect end users' premises to the IAPs' points of presence (POPs).
3. Backbone networks such as AT&T and UUNET that operate long-haul data networks and interconnect with one another through network access points (NAPs) and bilateral peering points to form the Internet backbone. IAPs connect to the backbone networks and pay them for bandwidth. Content publishers also pay the backbone networks for bandwidth consumed (either directly or indirectly through hosting services).
4. Content delivery networks (CDNs) such as Akamai that deliver content on behalf of content publishers using proprietary networks of caching servers.

Together, these players make up the digital supply chain for delivery of information goods on the Internet. The content publisher creates the content and networks help move the content.

Caches—the digital equivalent of warehouses—store and deliver the content to the users. Our study

Web caching refers to the temporary storage of content somewhere between the origin server and the end user to satisfy future requests for the same. The effectiveness of a cache is measured by the hit rate, which is the fraction of requests that are satisfied by the cache. Because IAP caches are located at points of aggregation, they are shared by a large number of users and hence demonstrate high hit rates.

Figure 1 The Internet Industry Structure



The **Internet industry structure** is a broad and complex network of various components and players involved in providing internet services and technologies. It can be broken down into several key segments:

1. Infrastructure Providers:

Network Operators: Companies that own and operate the physical networks that make up the internet, such as fiber optic cables, data centers, and cellular towers. Examples include AT&T, Verizon, and Comcast.

Data Centers: Facilities that house servers and other hardware for hosting websites, applications, and cloud services. Major players include Amazon Web Services (AWS), Google Cloud, and Microsoft Azure.

Content Delivery Networks (CDNs): Services that distribute and cache content closer to users to improve speed and performance, such as Akamai, Cloudflare, and Fastly.

2. Internet Service Providers (ISPs):

Residential ISPs: Companies that provide internet access to homes and small businesses. These can be broadband providers (e.g., Comcast, Charter) or DSL/fiber-optic providers.

Business ISPs: Providers that offer higher-speed and more reliable internet services tailored to businesses and enterprises.

3. Content Providers:

Websites and Online Services: Platforms and businesses that create and offer content or services online, including news sites, streaming services (Netflix, YouTube), and e-commerce sites (Amazon, eBay).

Social Media Platforms: Companies that provide social networking services and user-generated content, such as Facebook, Twitter, Instagram, and TikTok.

4. Application and Software Developers:
Web and Mobile App Developers: Companies and individuals who create applications and software for internet use, including everything from web browsers and productivity tools to mobile apps and games.
Software as a Service (SaaS): Providers offering software applications over the internet, typically on a subscription basis, such as Salesforce, Zoom, and Microsoft Office 365.

5. Regulatory Bodies and Standards Organizations:
Government Agencies: Bodies that regulate internet access, data privacy, and cyber security, like the Telecom Regulatory Authority of India (TRAI) in Bharat or the European Union's regulatory bodies.
Standards Organizations: Groups that develop and maintain technical standards for internet protocols and technologies, such as the Internet Engineering Task Force (IETF) and the World Wide Web Consortium (W3C).

6. Cybersecurity and Privacy Firms:
Security Providers: Companies that offer services and products related to internet security, such as firewalls, antivirus software, and intrusion detection systems. Examples include Norton, McAfee, and Palo Alto Networks.
Privacy Services: Providers that focus on protecting user privacy and data, including VPN services, encryption tools, and privacy-focused search engines.

7. Advertising and Marketing:
Ad Networks and Exchanges: Platforms that facilitate the buying and selling of online ads, such as Google Ads, Facebook Ads, and the Trade Desk.
Digital Marketing Agencies: Firms that help businesses with online marketing strategies, including SEO, content marketing, and social media management.
This structure is dynamic and continually evolving as technology advances and new business models emerge.